

Analysis of Contemporary Economic Issues

Guidance 6: Long-Term Investors and the Green Transition
Based on Philip Uschev's & Guntram Wolff's Working Paper

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ECONOMICS & MANAGEMENT**

Outline

- 1 Motivation & Research Question
- 2 Stylized Facts
- 3 Conceptual Framework
- 4 The Model
- 5 Key Predictions & Illustrations
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The Green Deal: How to Make It Work?

A Policy Question

How to make the Green Deal work properly?

A Research Question

Who **invests**/**does not invest** in green projects, and why?

This working paper (Ushchev & Wolff, 2026):

- Develops a model that rationalises key empirical patterns in green investment.
- Deals with the second question.

In a few words:

- We don't need to assume people are altruistic or have "green" preferences.
- Purely rational, selfish agents can end up investing in green bonds because those bonds act as insurance against climate catastrophe.
- Three factors drive who invests more: how patient you are, how wealthy you are, and how different agents are from each other.

Climate Change & the Global Tragedy of the Commons

Why do many countries/individuals invest too little?

- Climate protection is a **public good**: it means people/countries have an incentive to free ride.
- You can let others cut emissions while you keep emitting.
- **Global tragedy of the commons**: missing property rights for the remaining carbon budget $\implies$ **too much emission!**

Conceptual tradition:

- Hardin (1968): individually rational behaviour $\rightarrow$ collectively inefficient outcomes.
- Barrett (1994), Nordhaus (2015): public-goods models with free-riding incentives.
- Weitzman (2014): finite common resource systematically overused in non-cooperative settings.

Key Questions & Empirical Evidence

Do long-term holdings matter?

- Positive correlation of long-term investment with green investment.

Does economic size matter?

- Larger economies tend to invest more in green energy projects.

Does inequality matter?

- Positive correlation of cross-country inequality with global green investment.

This Lecture

A simple model that rationalises these three stylized facts.

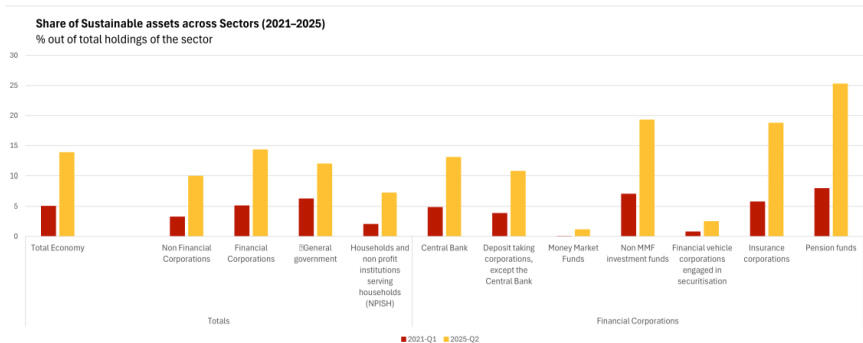
ECB's Securities Holdings Statistics (SHS) dataset:

- Quarterly data starting in 2021Q1.
- Breakdown by sector (at the country level).
- Breakdown by sustainability classification:
 - Green
 - Social
 - Sustainability
 - Sustainability-linked

Key observation: Euro-area sustainable debt share rose from **5.04%** (2021Q1) to **13.9%** (2025Q2), an increase of 8.8 percentage points.

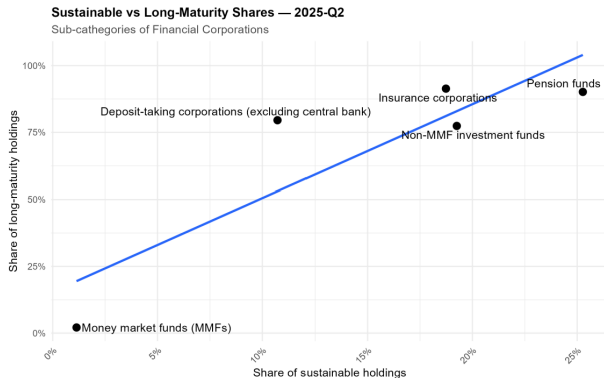
Sustainable debt share = portion of an entity's total financing allocated to initiatives with positive environmental, social, or sustainability outcomes

Share of Sustainable Assets Across Sectors



- **Sustainable Asset** = assets that are designed to be energy-efficient, repairable, and durable, and that support circular economy (e.g renewable energy infrastructure, funds with ESG ratings).
- Increase in sustainable debt holdings visible across **all sectors**.
- Financial corporations hold 91.2% of all debt securities.

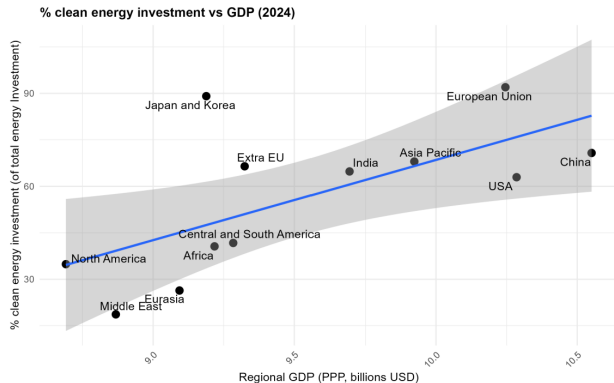
Stylized Fact (1): Long-Term Holdings Matter



Key finding:

- Positive correlation between share of long-maturity holdings and share of sustainable assets.
- **Pension funds** and **insurance corporations** (long-term investors): highest sustainable shares (~20–25%).
- **Money market funds** (short-term investors): near zero sustainable holdings.

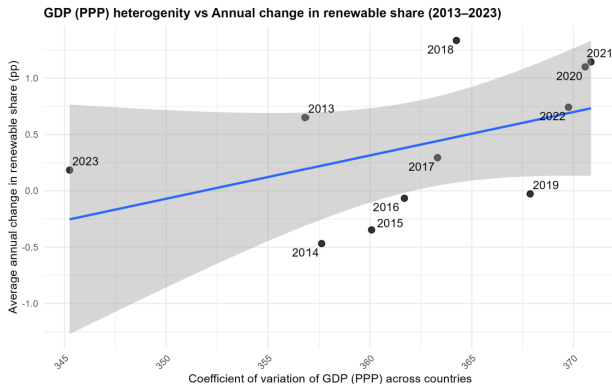
Stylized Fact (2): Size of the Economy Matters



Key finding:

- Larger economies tend to invest more in green energy projects.
- EU and Japan/Korea: high share of clean energy investment.
- Intuition: larger jurisdictions **internalise** a bigger fraction of the global benefit from its own green investment, so it has a stronger incentive to invest.

Stylized Fact (3): Inequality Matters



Key finding:

- More cross-country heterogeneity correlates with more green investment.
- Most surprising fact and the one the model explains most distinctively.

Note : These correlations are suggestive, not causal tests of the model. Many country-specific factors (policy ambition, energy endowments, political economy) also drive green investment.

Modelling Approach

- Strategic behaviour under **standard preferences**.
- No “green” or moral preferences, no warm-glow.

Key Modification

Two-period setup $\implies$ savings and investment decisions.

- **Time preference and wealth accumulation:** link financial decisions to green investments.
- **Novel mechanism:** heterogeneity in time preferences and wealth $\implies$ differences in contributing to climate protection (global public good).

Agents

Global economy: a world with n countries.

Each “agent” $i = 1, 2, \dots, n$ = the representative agent of a country. Each one has an initial wealth w_i and a discount factor β .

Heterogeneity in:

- Time preference (patience)
- Wealth/income

Time preference:

- Captured by the agent-specific **discount factor** β_i with $0 < \beta_i < 1$.
- Higher discount factor $\implies$ more patient agent $\implies$ more long-term oriented.
- More patience means more weight on future utility relative to present utility.

Investment Alternatives

Two types of assets:

- **Conventional bonds** (return r_b)
- **Green bonds** (return r_g)

Green bonds:

- Have **lower return** than conventional ($r_g < r_b$).
- Provide **insurance** against the risk of a major climate event.

Public Good

Global green investment $G = \sum_{j=1}^n g_j$ is a **public good**: it reduces the probability of an extreme climate event for **everyone**.

The Greenium

Definition

Greenium = differential between the returns on conventional and green investment:

$$r_b - r_g > 0.$$

Empirical evidence:

- Larcker and Watts (2020)
- Bolton and Kacperczyk (2021)
- Grzegorzczak and Wolff (2022)
- Zerbib (2019)

Returns treated as **exogenous** parameters. The stochastic element comes from the climate event risk.

Model Setup

The **World** has n countries, each represented by a single agent i . Each agent has initial wealth w_i and a personal discount factor β_i (between 0 and 1). A higher β means more patience (i.e. more weight on future utility relative to present utility).

There are **two periods**: present and future. In the present, the agent splits wealth into three uses: consumption now (c_i^0), conventional bond investment (b), and green bond investment (g). In the future, those investments pay returns r_b (conventional) and r_g (green), with $r_b > r_g$. The positive difference $r_b - r_g$ is the greenium, the return premium you sacrifice by going green.

Twist: There are two possible future states. With probability $p(G)$, the environment stays normal and all investments pay their standard returns. With probability $1 - p(G)$, an extreme climate event occurs and a fraction δ of all capital is destroyed. The probability $p(G)$ is increasing in aggregate global green investment G (the sum of all agents' green investments). So green investment is a public good: each euro of green investment by any agent raises $p(G)$ for everyone, reducing the risk of catastrophe.

Model Setup

- **Behaviour:** rational and selfish; each agent maximises lifetime expected utility.
- **Two time periods:** the present and the future.
- **Greenium:** $r_b - r_g > 0$.

Public good: aggregate green investment G .

- $p(G)$ = probability of **no** major climate event (increasing in G).
- $1 - p(G)$ = probability of the major climate event.
- Climate event $\implies$ a fraction $\delta \in (0, 1)$ of capital gets **destroyed**.

Model: Preferences

Agent i 's lifetime expected utility:

$$U_i = u(c_i^0) + \beta_i \mathbb{E}[u(z c_i^1) \mid G]$$

- β_i : agent i 's discount factor, $0 < \beta_i < 1$.
- z : share of assets which remains safe in the future:
 - No climate event $\implies z = 1$, with probability $p(G)$.
 - Climate event $\implies z = 1 - \delta$, with probability $1 - p(G)$.
- $\mathbb{E}[\cdot \mid G]$: expected value (probabilities conditional on G).

Conditioning on G : probabilities themselves are not fixed, they depend on how much green investment the world makes collectively.

Model: Budget Constraints

Each agent i faces two budget constraints:

Present:

$$w_i = c_i^0 + b_i + g_i$$

Future:

$$c_i^1 = (1 + r_b) b_i + (1 + r_g) g_i$$

Key variables:

- c_i^0 : present consumption
- c_i^1 : future consumption
- b_i : conventional investment
- g_i : green investment
- w_i : agent i 's initial wealth

Key Equations

1. Euler equation (present/future consumption trade-off):

$$u'(c_i^0) = \beta_i(1 + r_b) \mathbb{E} \left[z u'(z c_i^1) \mid G \right]$$

It captures the present vs future consumption trade-off: the marginal utility of consuming today equals the discounted expected marginal utility of consuming tomorrow (scaled by the gross return on conventional bonds).

2. Return/risk trade-off:

$$\underbrace{\frac{\partial \mathbb{E} [u(z c_i^1) \mid G] / \partial G}{\partial \mathbb{E} [u(z c_i^1) \mid G] / \partial c_i^1}}_{\text{MRS between } G \text{ and } c_i^1} = \underbrace{r_b - r_g}_{\text{greenium}}$$

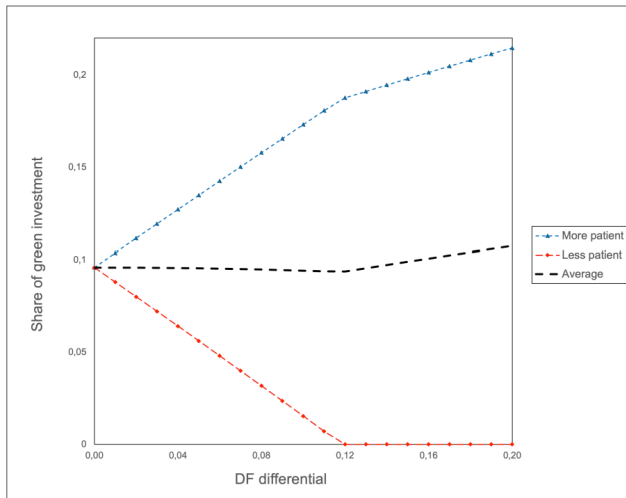
Intuition: The **marginal rate of substitution** between the insurance effect of green investment and future private consumption equals the **greenium**. Investors accept lower returns on green bonds up to the point where the insurance value of one more euro in green exactly compensates for the forgone return.

Key Predictions

The model makes three key predictions:

- ① **Long-term oriented investors** hold more green bonds (higher share of long-term investment $\leftrightarrow$ higher share of green bonds).
- ② **Wealthier investors** (larger economies) tend to invest more in green.
- ③ More **heterogeneity** in time preference and/or wealth leads to higher **aggregate** green investment.

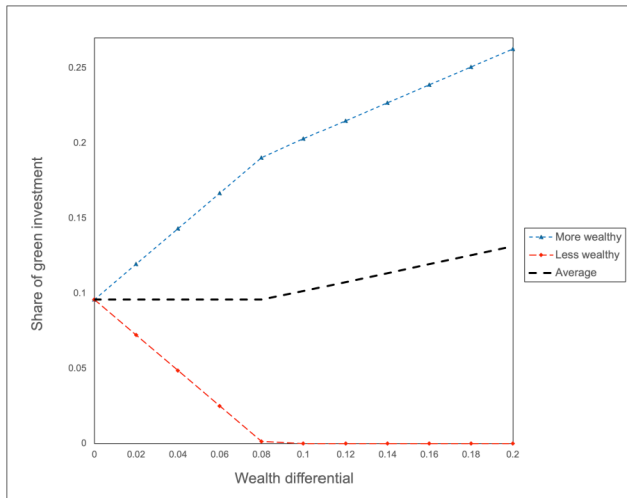
Illustration: The Role of Time Preference



Observations:

- More patient agent always holds a **greater share** of green assets.
- Share **increases** with the discount factor.
- Less patient agent's green investment eventually falls to **zero** (free-riding).
- Average green investment **rises** with heterogeneity.

Illustration: The Role of Wealth Distribution



Observations:

- Wealthier agents invest more into green.
- There exists a **threshold** of wealth below which green investment becomes zero.
- Increasing wealth dispersion → higher **aggregate** green investment.

Policy Implications

From the model's findings:

1. Role of patience and long-term investment:

- Policymakers cannot directly influence patience, but they can **facilitate long-term investment**.
- Review of **regulatory burdens** that long-term investors face when investing in green assets.
- Various regulations currently incentivise holding short-term government debt over long-term green assets.

2. Role of size and heterogeneity:

- Building coalitions via **climate clubs** should increase investment in climate mitigation.
- A larger club internalizes more of the public good benefit, and the heterogeneity within the club further boosts aggregate green investment.
- The EU exemplifies such a strong club for climate mitigation.

Conclusions

Main findings :

- Long-term oriented investors tend to invest more in green.
- So do wealthier investors/larger economies (size matters).
- More heterogeneity $\implies$ higher aggregate green investment.

Possible extensions:

- The model is deliberately simple (two periods, two states of the world) to deliver clean analytical results.
- This means it abstracts from many real-world complexities like policy uncertainty, technological change, or political economy constraints.

Bottom line: Green finance and investments can be strengthened through **long-term orientation** and **coalition building**.

Any Questions?